DINO-MTP: Topology-Aware Multimodal Fusion of GPS Trajectories and Optical Imagery for Remote Sensing Road Extraction

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***Abstract*—Accurate road extraction from remote sensing imagery is critical for urban mapping, navigation, and geospatial analysis, yet it remains highly challenging due to occlusions, thin and fragmented road structures, and the inherent limitations of single-modality data. To overcome these challenges, we propose topology-aware multimodal fusion (DINO-MTP), a novel end-to-end framework that synergistically fuses GPS trajectory data with optical imagery for robust multimodal road extraction. Leveraging a self-supervised DINOv3 backbone, our approach introduces topology-aware and morphology-aware multimodal feature learning to explicitly address cross-modality heterogeneity, structural discontinuities, and background clutter. The framework incorporates a bidirectional attention alignment with modular attention masking to achieve deep semantic alignment and effective noise suppression within a unified embedding space. In addition, it integrates spatial priors into channel attention to enhance the representation of elongated and interconnected road structures, while a feature-adaptive super-resolution pathway recovers fine details typically lost in conventional skip connections. An efficient dual-path cross-scale fusion module further ensures a coherent transition from global topology reconstruction to local geometric refinement. Extensive experiments and ablation studies on the BJRoad, RS-RGB-T, and Porto datasets demonstrate that DINO-MTP consistently surpasses state-of-the-art unimodal and multimodal methods, delivering clearer, more continuous, and topologically faithful road network extraction in complex urban environments.**

***Index Terms*—Multimodal fusion, remote sensing, road extraction, self-supervised representation, topology-aware.**

I. Introduction

RECENT advances in artificial intelligence (AI)-driven remote sensing have transformed image analysis from coarse scene-level interpretation to precise, pixelwise recovery of object geometry and topology [1]. This capability is particularly crucial for the extraction and graph-based modeling of elongated linear structures, such as roads, which underpin high-resolution semantic foundations for urban mapping, traffic analytics, and military intelligence [2]. As spatial resolution and acquisition frequency of remote sensing imagery continue to improve [3], the demand for highly accurate and robust segmentation methods has intensified to support large-scale road network mapping, disaster response, and land-cover dynamics assessment.

Although significant progress has been made in single-modality road extraction through advances in model architectures and attention mechanisms, such as end-to-end CNNs and Transformer-based methods, reliance on a single data source remains fundamentally limited [4], [5]. High-resolution optical imagery excels at capturing road boundaries and fine textures but is highly susceptible to shadows, specular reflections, and occlusions from vegetation or buildings, often resulting in discontinuities or false detections under challenging conditions [6]. Conversely, GPS trajectory data inherently capture road connectivity and hierarchical structure, accurately reflecting the traversable network, yet lack detailed information on road geometry, width, and contextual appearance [7]. These limitations have catalyzed the development of multimodal remote sensing semantic segmentation, particularly for linear-structure extraction tasks, such as roads, where integrating complementary data sources yields richer per-pixel feature representations and improved robustness in complex or heterogeneous environments [8]. Prior works have demonstrated the effectiveness of fusing imagery and trajectory data to enhance road connectivity under occlusions and low-contrast conditions [9]. Dual-branch and multiencoder architectures have been employed to separately capture texture and geometric cues, followed by multiscale fusion to improve recognition under varied shadows and occlusions [10], [11]. Trajectory-derived road probability maps have also been integrated with optical features to further improve segmentation continuity.

Concurrently, advances in self-supervised visual representation learning, most notably the DINO family [12], have enabled the extraction of structured semantic features from large-scale unlabeled data via teacher–student self-distillation, providing strong transferability to segmentation, detection, and clustering tasks. While initially applied to medical image segmentation for thin-structure delineation, these methods

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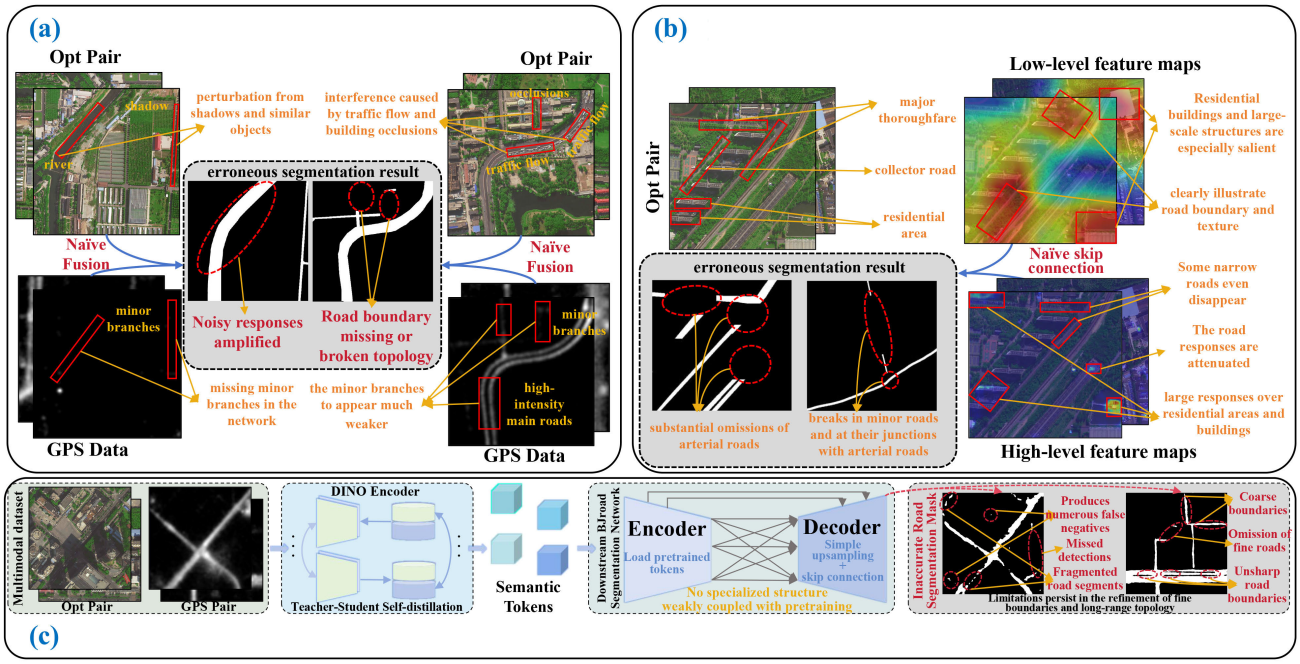


Fig. 1. Schematic of key challenges in self-supervised multimodal road extraction. (a) Cross-modal alignment and semantic drift caused by inconsistent modality properties. (b) Multiresolution topological reconstruction difficulties, leading to fragmented thin roads and oversmoothed regions. (c) Lack of unified end-to-end architectures, limiting fine-grained boundary recovery and topological integrity.

address similar challenges in remote sensing, including precise boundary localization and annotation scarcity. Recent works have extended self-supervised vision transformer pretraining to remote-sensing imagery, showing marked improvements for small objects and complex backgrounds [13], [14]. Furthermore, self-supervised and cross-domain learning approaches have demonstrated significant gains in multimodal segmentation tasks, particularly for road-related categories, even under limited annotation [15], [16]. Despite recent advances, as shown in Fig. 1, self-supervised multimodal road extraction still faces three fundamental challenges.

1) *Cross-Modal Alignment and Semantic Drift:* Systematic differences between modalities, such as optical imagery and GPS trajectories, lead to inconsistent geometric distortions and statistical properties. Naive fusion can amplify noise or suppress critical boundary cues, and most attention-based methods lack strictly bidirectional, explicit alignment. As shown in Fig. 1(a), this results in unreliable connectivity in extracted road regions.
2) *Multiresolution Topological Reconstruction:* Thin roads and large-area regions are represented differently at various network stages. Common fusion strategies rarely incorporate structural priors across scales, often leading to fragmented thin roads and oversmoothed small targets. As shown in Fig. 1(b), this undermines accurate reconstruction of complex road topology.
3) *Unified End-to-End Learning and Data Scarcity:* Large-scale, densely labeled multimodal datasets are expensive, and current pipelines often focus on encoder pretraining while neglecting holistic encoder–decoder optimization. As shown in Fig. 1(c), without adaptive cross-scale fusion, it remains difficult to recover fine boundaries and ensure topological integrity under limited supervision.

To address these challenges, we introduce topology-aware multimodal fusion (DINO-MTP), a unified end-to-end framework for multimodal remote sensing road extraction that synergistically fuses GPS trajectory data with optical imagery. Built on the self-supervised DINOv3 backbone [17], DINO-MTP employs the proposed MDFFM module for accurate semantic alignment and noise suppression of cross-modal features via bidirectional attention alignment and modular attention masks (MAMs), thereby addressing semantic drift caused by modal heterogeneity. The SPMCA module embeds multiscale spatial priors to enhance the representation of slender road structures and alleviate fragmentation in multiscale topological reconstruction. The FAPMM and DPCF-ECA modules construct an end-to-end cross-scale fusion decoding path, enabling collaborative reconstruction of fine road boundaries and topological integrity under limited supervision. Our main contributions are as follows.

1) We propose a multimodal data feature fusion module (MDFFM), featuring a novel duplex attention alignment (DAA) for symmetric, token-level cross-modal alignment, and an MAM for explicit noise suppression and selective enhancement of semantically consistent regions.
2) We design a spatial-prior-modulated channel attention (SPMCA) that explicitly embeds multiscale spatial priors into the channel attention process, preserving sensitivity to elongated and thin road structures while enhancing discriminative representation across scales.
3) We introduce a feature adaptive prior modulation mechanism (FAPMM) and a dual-path cross-fusion

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with efficient channel attention (DPCF-ECA) decoder, enabling high-fidelity reconstruction of fine road details and robust integration of global and local semantic cues for continuous road network extraction.

The article is structured as follows. Section II provides an overview of the related works. Section III introduces the proposed DINO-MTP framework and its associated components. The experimental results and discussion analysis are presented in Section IV. At last, the conclusion is drawn in Section V.

II. Related Works

A. Single-Modal Remote Sensing Road Extraction

Extracting road networks from remote sensing imagery using a single modality presents considerable challenges due to complex backgrounds, variable illumination, and occlusions. Progress in this field has been driven by innovations in both global context modeling and local structure preservation. Transformer-based frameworks have expanded the capacity for multiscale feature extraction and boundary delineation in high-resolution scenarios [18]. The introduction of sparse self-attention and feature alignment modules has mitigated feature shift and texture confusion, thereby strengthening single-modal representations in challenging scenes [19]. To cope with the computational demands of ultrahigh-resolution inputs, score-map-driven fast fusion mechanisms have been developed, resulting in reduced redundancy and accelerated inference [20]. State-space models (SSMs) have shown effectiveness in capturing long-range dependencies for large-scale images, enabling more scalable and efficient semantic segmentation [21]. Integrating SSM-based encoders with lightweight decoders further enhances the ability to parse complex spatial structures while maintaining computational efficiency [22]. Addressing challenges such as annotation scarcity and overfitting, transfer learning has been incorporated into enhanced U-Net variants, leading to improved boundary accuracy and generalization in high-resolution segmentation [23]. For hyperspectral data, dual-branch architectures that jointly process spatial and spectral information have demonstrated robust performance by effectively modeling their intrinsic coupling [24]. Despite these advances, from CNNs to Transformer- and SSM-based models, single-modal approaches are still hindered by illumination changes, texture ambiguities, and occlusion effects, underlining the potential of multimodal strategies.

B. Multimodal Remote Sensing Road Extraction

Integrating multiple remote sensing modalities has become an effective strategy for enhancing the robustness and generalizability of road extraction models, particularly in heterogeneous and challenging environments. Recent studies have focused on designing architectures and fusion mechanisms that efficiently exploit complementary information from diverse data sources. Lightweight multibranch Transformer networks have been developed to address modality discrepancy and enable efficient cross-modal feature sharing [25]. To bridge the substantial spectral and structural differences between optical and SAR data, cross-modal enhancement modules have been introduced for effective feature alignment, yielding strong land-cover classification performance [26]. In the context of RGB–nDSM data, mid-level Transformer fusion modules have been proposed to better leverage geometric and spectral complementarity, thereby improving the joint modeling of geometry and appearance cues [10]. For dynamic scenarios involving multitemporal imagery, spatiotemporal fusion frameworks based on Transformers have shown effectiveness in integrating multisource time series with ultrahigh-resolution data, resulting in improved stability under significant temporal variations [9]. Hybrid architectures combining CNNs and Transformers enable joint modeling of global and local structures for enhanced spatial–spectral fusion [27]. Specifically targeting road extraction, multiscale attention fusion has been employed to exploit the complementary strengths of optical imagery and DSM, improving the recognition of thin road structures and maintaining topological connectivity [28]. Despite these advancements, challenges remain in fully exploiting complementary modalities, aligning heterogeneous features, and maintaining stability in diverse environments.

C. DINO for Remote Sensing Image Processing

The DINO framework leverages a teacher–student self-distillation approach to learn structured image representations without relying on manual annotations, making it highly suitable for remote sensing tasks [12]. Recent advances have extended DINO and related self-supervised paradigms to address key challenges in multimodal integration, weak supervision, and cross-domain generalization within remote sensing contexts. To tackle the challenge of insufficient discriminative feature learning, Ding et al. [29] proposed SLCGC, a lightweight self-supervised low-pass contrastive graph clustering network, which considerably boosts the clustering accuracy and noise robustness. Furthermore, to mitigate the issues of weak robustness, limited adaptability, and insufficient discriminative feature learning in unsupervised hyperspectral image clustering, Ding et al. [30] introduced the S^2LGCC method, which provides a valuable technical breakthrough for unsupervised visual representation learning in remote sensing. To address the challenges of graph structure mining and dynamic updating in deep graph clustering for hyperspectral imagery, Ding et al. [31] proposed the AHSGC method, which achieves state-of-the-art clustering accuracy and superior computational efficiency in unsupervised remote sensing representation learning. To capture spatial–spectral coupling and sequential features effectively, Hong et al. [32] proposed SpectralGPT, the first spectral remote sensing foundation model, which achieves state-of-the-art performance on multiple downstream tasks and provides an efficient paradigm for spectral big data processing. To address the challenge of heterogeneous input encoding, PerceiverIO has been introduced as a general-purpose multimodal Transformer, unifying cross-modal input representations and facilitating the transfer of self-supervised features to segmentation tasks [33]. In generative self-supervision, diffusion-based pretraining strategies, such as RS-Dseg, have shown promise in enhancing high-resolution feature learning under weak supervision, leading to improved segmentation accuracy and detail preservation [34]. Sparse multimodal vision transformers further combine self-supervised and weakly supervised signals to enable effective feature transfer and expression across diverse remote sensing

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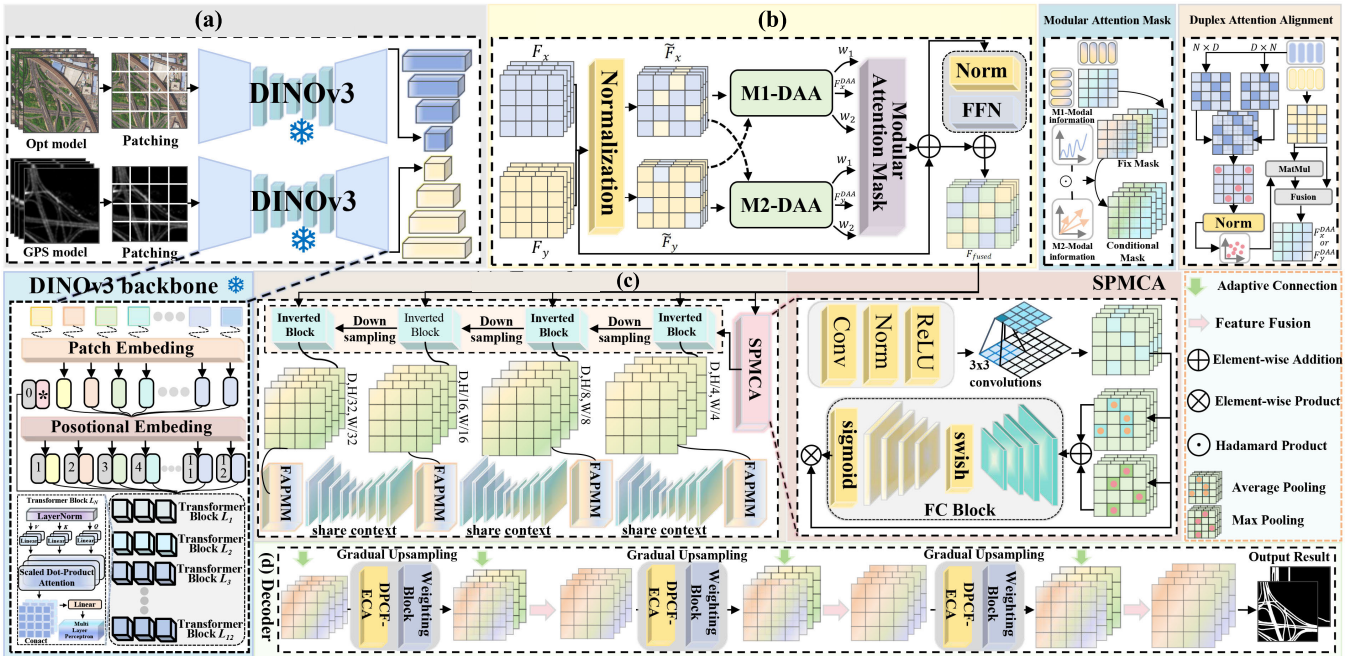


Fig. 2. Overview of the proposed DINO-MTP framework. The architecture integrates dual-stream DINOv3 backbones for multimodal feature extraction, a feature fusion module with DAA for cross-modal semantic integration, and a hierarchical encoder–decoder with adaptive enhancement modules for progressive and topologically consistent road extraction from remote sensing imagery. (a) Input multimodal data. (b) MDFFM. (c) Encoder.

modalities [35]. Robustness to seasonal and regional domain shifts has been improved by integrating multiple teacher signals, including DINO within the SeCo self-supervised framework, which significantly boosts cross-domain generalization for scene classification and segmentation [36]. In addition, SimDINO introduces coding-rate regularization during self-supervised training to mitigate overfitting and enhance the transferability and robustness of dense prediction models in complex and cross-domain remote sensing scenarios [37]. The application of DINO and self-supervised learning in remote sensing is evolving toward multimodal fusion and robust cross-domain transfer.

III. Methodology

A. Overall Architecture

As shown in Fig. 2, the DINO-MTP framework adopts a unified multimodal pipeline for robust road extraction. The input comprises rasterized GPS trajectories I_x and optical imagery I_y, which are independently encoded by dual DINOv3 backbones to obtain modality-specific features F_x and F_y. These unimodal features are fused via the MDFFM, incorporating DAA and MAM, to produce a semantically aligned multimodal representation F_{fused}. Subsequently, F_{fused} is hierarchically refined by the encoder, which integrates SPMCA. This yields multiscale encoder features $F_{\text{enc}}^{(i)}$ that capture both fine structural details and global context. In the decoding stage, high-level encoder outputs and channel-enhanced features are combined through skip connections and further processed by the FAPMM and the DPCF-ECA, resulting in decoder features $F_{\text{dec}}^{(i)}$ that progressively reconstruct the road network. Finally, the aggregated decoder output F_{out} is mapped by a convolutional prediction head to generate the final road mask F_{pred}. This architecture enables comprehensive multimodal representation learning and ensures accurate, topologically consistent road extraction. The overall calculation process of DINO-MTP is shown in Algorithm 1.

B. DINOv3 Backbone

As depicted in Fig. 2(a), the DINOv3 backbone is employed as a modality-agnostic high-dimensional feature extractor within our framework. It is trained under a self-supervised paradigm to capture unified and semantically meaningful visual representations from large-scale multimodal data, specifically rasterized GPS trajectories I_x and optical imagery I_y. By leveraging self-distillation and extensive pretraining, DINOv3 encodes discriminative and transferable tokens that offer substantial advantages for road extraction. Formally, for each input image $I_m \in \mathbb{R}^{H\times W\times 3}$ from modality $m \in \{x, y\}$, DINOv3 first partitions the image into nonoverlapping patches of size $P \times P$, yielding $N = HW/P^2$ patches. Each patch is then flattened and mapped to a D-dimensional embedding via a linear projection

$$Z_0^{(m)} = P(I_m)\, E_{\text{lin}} + E_{\text{pos}} \tag{1}$$

where $P(\cdot)$ denotes the patch extraction and flattening operation, $E_{\text{lin}} \in \mathbb{R}^{(P^2\cdot 3)\times D}$ is the learnable linear projection matrix, and $E_{\text{pos}} \in \mathbb{R}^{N\times D}$ represents the positional embedding, which encodes spatial order information within the patch sequence. The resulting embedded tokens are sequentially processed by a stack of Lth Transformer layers, each comprising multihead self-attention and feed-forward submodules, yielding hierarchical feature refinement

$$Z_l^{(m)} = \text{TransLayer}_l\left(Z_{l-1}^{(m)}\right), \quad l = 1, 2, \ldots, L \tag{2}$$

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Algorithm 1 Overall Workflow of DINO-MTP

Input: Rasterized GPS trajectory I_x, optical image I_y
Output: Road segmentation mask F_{pred}

Step 1: Multimodal Feature Extraction
$F_x \leftarrow \text{DINOv3}(I_x)$; // For GPS
$F_y \leftarrow \text{DINOv3}(I_y)$; // For Optical
Step 2: Feature Fusion and Alignment
$F_{\text{fused}} \leftarrow \text{MDFFM}(F_x, F_y)$; // Feature Fusion
Step 3: Hierarchical Encoding
for $i = 1$ **to** L **do**
  $F_{\text{enc}}^{(i)} \leftarrow \text{SPMCA}(F_{\text{fused}})$; // Spatial-channel modulation
  $F_{\text{enc}}^{(i)} \leftarrow \text{FAPMM}(F_{\text{enc}}^{(i)})$; // Adaptive prior enhancement
Step 4: Hierarchical Decoding with Feature Fusion
for $i = L$ **to** 1 **do**
  $F_{\text{dec}}^{(i)} \leftarrow \text{Fuse}\big(\text{Up}(F_{\text{enc}}^{(i+1)}), F_{\text{enc}}^{(i)}\big)$; // Skip connection fusion
  $F_{\text{dec}}^{(i)} \leftarrow \text{DPCF-ECA}(F_{\text{dec}}^{(i)})$; // Dual-path context fusion
Step 5: Prediction
$F_{\text{out}} \leftarrow F_{\text{dec}}^{(1)}$; // Final decoder output
$F_{\text{pred}} \leftarrow \text{Conv}_{\text{out}}(F_{\text{out}})$ **return** F_{pred}

where $\text{TransLayer}_l(\cdot)$ denotes the lth Transformer block. After L layers, the final modality-specific feature sequence is obtained as follows:

$$F_m = Z_L^{(m)} = \text{DINOv3}\left(I_m\right), \quad m \in \{x, y\}. \tag{3}$$

The extracted features F_x and F_y, representing unified and high-level visual abstractions of the GPS trajectory and optical branches, respectively, serve as the input representations to the subsequent multimodal fusion and alignment module.

C. Multimodal Data Feature Fusion Module

As illustrated in Fig. 2(b), the MDFFM is designed to effectively integrate and align heterogeneous information from rasterized GPS trajectories and optical imagery within a unified feature space. GPS trajectories provide globally coherent but spatially sparse priors for road localization, while optical images contribute rich texture and precise boundary details. However, significant modality discrepancies, in terms of spatial distribution, semantic granularity, and noise characteristics, make naive fusion suboptimal, often leading to semantic misalignment and error propagation. To address this, we propose a cooperative fusion architecture based on DAA and MAM, which jointly enforce bidirectional semantic correspondence and adaptive noise suppression, yielding robust cross-modal features F_{fused} tailored for road extraction. Given the bimodal features $F_x = \text{DINOv3}(I_x)$ and $F_y = \text{DINOv3}(I_y)$, we first perform joint feature normalization to eliminate distributional biases

$$\left[\widetilde{F}_x, \widetilde{F}_y\right] = \frac{\left[F_x, F_y\right] - \mu\left(\left[F_x, F_y\right]\right)}{\sigma\left(\left[F_x, F_y\right]\right)} \tag{4}$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ denote the mean and standard deviation computed over the concatenated features. The normalized representations are then input to parallel DAA branches, where cross-modal attention is established as follows:

$$\left[F_x^{\text{DAA}}, F_y^{\text{DAA}}\right] = \mathcal{A}_{\text{duplex}}\left(\left[\widetilde{F}_x, \widetilde{F}_y\right]; \theta_{\text{DAA}}\right) \tag{5}$$

where $\mathcal{A}_{\text{duplex}}(\cdot)$ is the DAA operator parameterized by θ_{DAA}, responsible for bidirectional semantic alignment. The DAA outputs are subsequently processed by MAM, which dynamically computes a relevance-based mask to selectively aggregate semantically consistent regions and suppress modality-specific noise as follows:

$$F_{\text{mask}} = \mathcal{M}\left(F_x^{\text{DAA}}, F_y^{\text{DAA}}; \theta_{\text{MAM}}\right) \tag{6}$$

where $\mathcal{M}(\cdot)$ is the MAM function parameterized by θ_{MAM}. Finally, a residual feedforward transformation further enhances the fused features as follows:

$$F_{\text{fused}} = \text{FFN}\left(\text{Norm}\left(F_{\text{mask}}\right)\right) + F_{\text{mask}}. \tag{7}$$

The resulting F_{fused} is fed into subsequent encoding and prior modeling modules for hierarchical refinement.

1) Duplex Attention Alignment: As the core of MDFFM, DAA is dedicated to learning symmetric, bidirectional attention mappings between modalities. Different from conventional unidirectional cross-attention, DAA enables both "GPS attending to optical" and "optical attending to GPS," thus capturing complementary structures and semantic interactions. Formally, for input features of size $N \times D$, we compute

$$\left(Q_x, K_y, V_y\right) = \mathcal{L}\left(\widetilde{F}_x, \widetilde{F}_y; W_Q, W_K, W_V\right) \tag{8}$$

where $\mathcal{L}(\cdot)$ represents linear projection with learnable matrices W_Q, W_K, and W_V. The bidirectional attention is formulated as follows:

$$A_{x \leftarrow y} = \text{Softmax}\left(\frac{Q_x K_y^{\mathsf{T}}}{\sqrt{d_k}}\right) V_y \tag{9}$$

$$A_{y \leftarrow x} = \text{Softmax}\left(\frac{Q_y K_x^{\mathsf{T}}}{\sqrt{d_k}}\right) V_x \tag{10}$$

where d_k is the dimensionality of the key vectors. The fused DAA features are then obtained by concatenation

$$F_{\text{DAA}} = A_{x \leftarrow y} \oplus A_{y \leftarrow x} \tag{11}$$

with $\oplus$ denoting feature concatenation. This symmetric structure enables robust information exchange and alignment.

2) Modular Attention Mask: The MAM further enhances cross-modal fusion by generating a data-adaptive mask that filters out modality-unique noise while reinforcing consistent road cues. The mask is computed as follows:

$$\mathcal{M} = \sigma\left(W_2\, \delta\left(W_1\left[F_x^{\text{DAA}} \,\|\, F_y^{\text{DAA}}\right]\right)\right) \tag{12}$$

where $[\cdot \,\|\, \cdot]$ denotes concatenation, $\delta(\cdot)$ is ReLU activation, and $\sigma(\cdot)$ is the sigmoid function. The mask is then applied via elementwise multiplication as follows:

$$F_{\text{mask}} = \mathcal{M} \odot F_{\text{DAA}} \tag{13}$$

where $\odot$ is the Hadamard product. This operation adaptively suppresses irrelevant background and noisy artifacts, while

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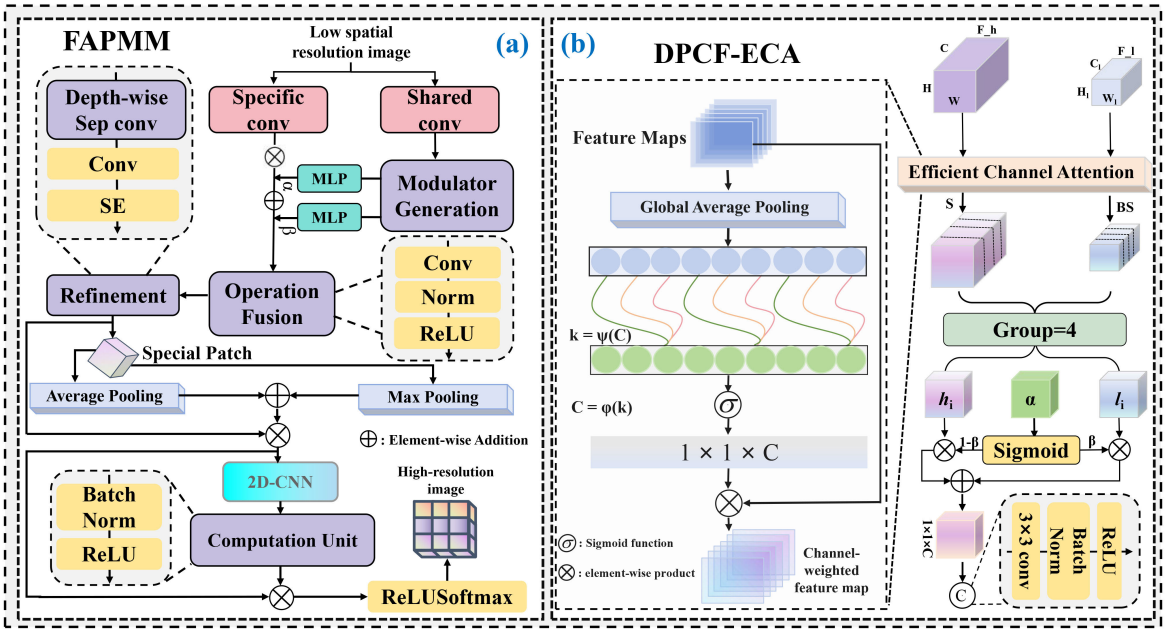


Fig. 3. Schematic of the proposed modules. (a) FAPMM adaptively integrates spatial and semantic features through patch-level mixing and operation fusion. (b) DPCF-ECA performs dual-path, groupwise feature fusion with ECA, enhancing both fine detail and global structural consistency.

highlighting spatially continuous and semantically confirmed road regions. The resulting features thus achieve robust cross-modal filtering and saliency enhancement.

D. Spatial-Prior-Modulated Channel Attention

As illustrated in Fig. 2(c), the encoding stage of multimodal remote sensing data is challenged by pronounced spatial and channelwise heterogeneity, rasterized GPS trajectories exhibit sampling artifacts and incomplete coverage, while optical imagery contains complex background textures, such as shadows, vehicles, and occlusions. To address these challenges, we propose an SPMCA at the early encoding stage. The SPMCA is specifically designed to enhance the global importance of multimodal features along the channel dimension, while integrating spatial saliency constraints via convolutional priors. This joint modeling reinforces sensitivity to road boundaries and maintains the connectivity of intricate road regions. Given an input feature map $F \in \mathbb{R}^{C\times H\times W}$, SPMCA first generates an intermediate spatial feature using a 1×1 convolution followed by normalization and nonlinear activation

$$F' = \mathrm{ReLU}\left(\mathrm{Norm}\left(\mathrm{Conv}_{1\times 1}(F)\right)\right). \tag{14}$$

A subsequent 3×3 convolution is then used to capture multiscale semantic patterns as follows:

$$F_{\mathrm{ms}} = \mathrm{Conv}_{3\times 3}\left(F'\right). \tag{15}$$

This produces a feature map F_{ms} that encodes both local spatial details and global structural information. To extract global and local channel statistics, we perform both average and max pooling along the spatial dimensions and concatenate

$$Z = \mathrm{Concat}\left(\mathrm{AvgPool}(F_{\mathrm{ms}}), \mathrm{MaxPool}(F_{\mathrm{ms}})\right) \tag{16}$$

where $\mathrm{AvgPool}(F_{\mathrm{ms}}), \mathrm{MaxPool}(F_{\mathrm{ms}}) \in \mathbb{R}^{C\times 1\times 1}$. The aggregated descriptor $\mathbf{Z}$ is then projected by a two-layer fully connected network, interleaved with normalization and activation, to produce channel attention weights $\mathbf{S}$

$$S = \sigma\left(W_2\,\delta\left(\mathrm{BN}\left(W_1\,\mathrm{LN}\left(\mathbf{z}\right)+\mathbf{b}_1\right)\right)+\mathbf{b}_2\right) \tag{17}$$

where $W_1 \in \mathbb{R}^{C/r\times C}$ and $W_2 \in \mathbb{R}^{C\times C/r}$ are the weights of the two fully connected layers, r is the channel reduction ratio, LN and BN denote layer and batch normalization, $\delta(\cdot)$ is the ReLU activation, and $\sigma(\cdot)$ is the Sigmoid function. Finally, the channel attention weights are applied to F_{ms} with a residual modulation term to emphasize discriminative regions and promote feature stability as follows:

$$P = F_{\mathrm{ms}} \oplus \lambda\left(F_{\mathrm{ms}} \odot S - F_{\mathrm{ms}}\right) \tag{18}$$

where $\odot$ and $\oplus$ denote elementwise multiplication and addition, respectively, and $\lambda \in [0,1]$ is a residual scaling hyperparameter. The refined feature P is subsequently fed into the inverted block (IB) interaction layers, where it interacts with the fused multimodal feature F_{fused} and is progressively downsampled to form a multilevel feature pyramid, facilitating topologically coherent road network extraction.

E. Feature Adaptive Prior Modulation Mechanism

In remote sensing scenarios, road networks are predominantly composed of slender, elongated structures whose width is often close to the image pixel scale. Consequently, low-resolution features tend to result in blurred road boundaries, fractured thin roads, and missing intersections. To address these challenges and enhance the recovery of fine spatial details and road connectivity, we propose an FAPMM, as illustrated in Fig. 3(a). This mechanism fuses adaptive prior modulation with lightweight structural refinement, enabling joint optimization across global semantics and local geometry.

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Given the downsampled low-resolution features $X_{\mathrm{LR}} = E_l^{\downarrow} \in \mathbb{R}^{C_l \times H_l \times W_l}$, we first apply two parallel branches

$$F_{\mathrm{spec}} = W_{\mathrm{spec}} * X_{\mathrm{LR}} + b_{\mathrm{spec}} \tag{19}$$

$$F_{\mathrm{share}} = W_{\mathrm{share}} * X_{\mathrm{LR}} + b_{\mathrm{share}} \tag{20}$$

where $*$ denotes convolution, and $W_{\mathrm{spec}}, W_{\mathrm{share}} \in \mathbb{R}^{C_l \times C_l \times k \times k}$. To extract the global context, global average pooling is conducted on the shared branch

$$\mathbf{G} = \mathrm{GAP}\left(F_{\mathrm{share}}\right) = \frac{1}{H_l W_l} \sum_{j=1}^{H_l} \sum_{k=1}^{W_l} F_{\mathrm{share}}(:, j, k) \tag{21}$$

where $\mathbf{G} \in \mathbb{R}^{C_l}$ encodes the channelwise context. The global descriptor $\mathbf{z}$ is then passed through two lightweight MLPs to generate channelwise scaling ($\boldsymbol{\alpha}$) and bias ($\boldsymbol{\beta}$) terms

$$\boldsymbol{\alpha} = \sigma\left(W_{\alpha 2}\, \delta\left(W_{\alpha 1}\mathbf{z}\right)\right), \quad \boldsymbol{\beta} = \tanh\left(W_{\beta 2}\, \delta\left(W_{\beta 1}\mathbf{z}\right)\right) \tag{22}$$

where $W_{\alpha 1}, W_{\alpha 2}, W_{\beta 1}$, and $W_{\beta 2}$ are learnable weights, $\delta(\cdot)$ is ReLU, and $\sigma(\cdot)$ and $\tanh(\cdot)$ are Sigmoid and Tanh activations, respectively. $\boldsymbol{\alpha}, \boldsymbol{\beta} \in \mathbb{R}^{C_l}$ are broadcast spatially to $M_\alpha, M_\beta \in \mathbb{R}^{C_l \times H_l \times W_l}$. The channelwise affine modulation is performed

$$F_{\mathrm{mod}} = M_\alpha \odot F_{\mathrm{spec}} + M_\beta \tag{23}$$

where $\odot$ is elementwise multiplication. This step adaptively recalibrates features using global priors, enhancing discriminative patterns, and suppressing noise. To aggregate local spatial cues, we introduce a fusion operator

$$F_{\mathrm{fuse}} = \mathrm{ReLU}\left(\mathrm{Norm}\left(W_o * F_{\mathrm{mod}} + b_o\right)\right) \tag{24}$$

where W_o is a convolution kernel and $\mathrm{Norm}(\cdot)$ denotes normalization. This encourages each feature to aggregate responses from its local neighborhood, ensuring contextual continuity. For fine-grained enhancement, a refinement block is employed. First, a depthwise separable convolution extracts detailed textures as F_{dw}, followed by a channel recalibration:

$$F_{\mathrm{ref}} = \left(\mathrm{R}\left[\sigma\left(W_2\, \delta\left(W_1\, \mathrm{GAP}\left(F_{\mathrm{dw}}\right)\right)\right), H_l, W_l\right]\right) \odot F_{\mathrm{dw}} \tag{25}$$

where W_1 and W_2 are fully connected weights, and $\mathrm{R}[\cdot, H_l, W_l]$ denotes broadcasting. The refined feature F_{ref} is partitioned into patches $F_{\mathrm{patch}} \in \mathbb{R}^{C_p \times H_p \times W_p}$ for localized enhancement. The average pooling and max pooling are applied and fused

$$F_{\mathrm{pool}} = \mathrm{AvgPool}\left(F_{\mathrm{patch}}\right) \oplus \mathrm{MaxPool}\left(F_{\mathrm{patch}}\right) \tag{26}$$

where $\oplus$ indicates elementwise addition. After two residual connections, a 2-D convolution, batch normalization, and ReLU–Softmax activation, the high-resolution prediction is as follows:

$$I_{\mathrm{HR}} = \mathrm{Softmax}\left(\mathrm{ReLU}\left(R_1 + \mathrm{BN}\left(\left(F_{\mathrm{ref}} + F_{\mathrm{pool}}\right)\right)\right)\right) \tag{27}$$

where BN is batch normalization, and R_1 is the residual input. The final output I_{HR} delivers high geometric fidelity with sharp road boundaries, slender segments, and intersection structures, providing robust feature support for the subsequent decoding and topologically coherent road reconstruction.

F. Dual-Path Cross-Fusion With ECA

As shown in Fig. 3(b), to enable the decoder to effectively integrate spatial and semantic features across multiple scales, particularly to simultaneously highlight fine-grained minor roads and robustly preserve trunk roads, we propose a DPCF-ECA. This design extends the ECA [38] mechanism with a novel, learnable balancing coefficient α, which adaptively reconciles the contributions of high-resolution features (F_h) and low-resolution features (F_l). Given a pair of features $F_h, F_l \in \mathbb{R}^{C \times H \times W}$ from different resolutions, we first enhance channelwise dependencies using a shared ECA block as follows:

$$\hat{F}_h = \mathcal{A}_{\mathrm{ECA}}\left(F_h\right), \quad \tilde{F}_l = \mathcal{A}_{\mathrm{ECA}}\left(F_l\right) \tag{28}$$

where $\mathcal{A}_{\mathrm{ECA}}(\cdot)$ denotes the ECA operation. The resulting $\hat{F}_h$ and $\tilde{F}_l$ suppress irrelevant background and highlight road-relevant structures. To prevent subtle road features from being overwhelmed by global semantics, we partition the channels into $G = 4$ groups as follows:

$$\widetilde{F}_h = \left[h_1, h_2, h_3, h_4\right], \quad \widetilde{F}_l = \left[l_1, l_2, l_3, l_4\right] \tag{29}$$

where $h_i, l_i \in \mathbb{R}^{C/4 \times H \times W}$. This groupwise strategy allows fine-grained, scale-aware fusion within subspaces. We then introduce a learnable parameter α to adaptively balance the fusion of details and context. After mapping α via a Sigmoid function, the balancing coefficient β is broadcast to the spatial size of each group as follows:

$$\beta = \sigma\left(\alpha\right), \qquad \beta \in (0,1)^{\frac{C}{4} \times H \times W} \tag{30}$$

where a larger β prioritizes high-resolution features (for recovering broken or missing thin roads), and a smaller β enhances low-resolution context (to avoid oversharpening). For each group, cross-scale fusion is performed as follows:

$$f_i = \beta \odot h_i + (1 - \beta) \odot l_i \tag{31}$$

where $\odot$ is elementwise multiplication. This soft reweighting adaptively merges representations from both resolutions within each group, effectively decoupling and recombining multiscale features. The fused groups are calculated as follows:

$$F_{\mathrm{out}}^{h} = F_h + \mathrm{ReLU}\left(\mathrm{BN}\left(W_3 * \mathrm{C}\left(f_1, f_2, f_3, f_4\right) + b\right)\right) \tag{32}$$

where W_3 is a 3×3 convolution kernel, and $\mathrm{C}(\cdot)$ is channelwise concatenation. The residual connection preserves the original information while the fusion block enhances structural continuity and semantic stability. The resulting F_{out}^{h} provides high-quality, scale-consistent features for subsequent decoding and progressive road network reconstruction.

G. Loss Function

To address the challenges of severe class imbalance, varying region importance, and the preservation of thin road boundaries and topological connectivity in complex remote sensing scenes, we formulate a composite loss function that integrates the weighted cross-entropy (WCE), generalized dice loss (GDL), and boundary-aware loss (BAL). Given the predicted probability map $\mathbf{P} \in \mathbb{R}^{H \times W \times C}$ and the label $\mathbf{Y}$, we first employ

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WCE to alleviate the dominance of background pixels and to enhance sensitivity to the minority road class

$$\mathcal{L}_{\mathrm{WCE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} w_c\, Y_{i,c} \log P_{i,c} \tag{33}$$

where $w_c = 1/\log(\kappa + f_c)$ is the class-rebalancing weight dependent on class frequency f_c and smoothing hyperparameter $\kappa > 1$. This term helps maintain discrimination for underrepresented road pixels. To promote the structural integrity of thin roads, we incorporate the GDL as follows:

$$\mathcal{L}_{\mathrm{GDL}} = 1 - \sum_{c=1}^{C} \zeta_c \,\mathrm{Dice}_c \tag{34}$$

with class weights $\zeta_c = 1/\left(\sum_{i=1}^{N} Y_{i,c}\right)^2$. This formulation directly maximizes region overlap while amplifying the contribution of minority classes, which is particularly effective for maintaining connectivity and continuity in narrow road segments. Recognizing the limitations of region-based losses in refining boundary localization, we further introduce BAL

$$\mathcal{L}_{\mathrm{BAL}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k\in\{0,1\}} w_k\, \mathbb{I}\left\{B_i^{\mathrm{gt}} = k\right\} \log B^{\mathrm{p}} i \tag{35}$$

where $B^{\mathrm{gt}}i$ and $B^{\mathrm{p}}i$ denote the ground-truth and predicted boundary probabilities for pixel i, respectively. η controls the relative importance of boundary ($k = 1$) and nonboundary ($k = 0$) pixels. $\mathbb{I}\cdot$ is the indicator function. The final objective combines the complementary strengths of these losses

$$\mathcal{L}_{\mathrm{total}} = \lambda_1 \mathcal{L}_{\mathrm{WCE}} + \lambda_2 \mathcal{L}_{\mathrm{GDL}} + \lambda_3 \mathcal{L}_{\mathrm{BAL}} \tag{36}$$

where $\lambda_1, \lambda_2, \lambda_3 \geq 0$ are balancing coefficients. This collaborative loss design enables the network to jointly address class imbalance, structural consistency, and boundary accuracy. It simultaneously drives the decoder to distinguish road regions from background, recover multiscale road morphology, and reconstruct high-frequency edge details, thus significantly improving the accuracy, completeness, and topological faithfulness of road extraction in remote sensing scenarios.

IV. Experiments and Analysis

A. Dataset Description

To rigorously evaluate the accuracy and topological robustness of the proposed DINO-MTP framework under diverse urban scenarios, we conduct experiments on three complementary multimodal benchmarks: BJRoad, Porto, and RS-RGB-T. These datasets encompass a broad spectrum of sensing modalities and urban scene complexities, thereby providing a comprehensive basis for assessing road-extraction performance.

1) BJRoad Dataset [53]: This dataset consists of paired optical and trajectory-based modalities, with the latter being relatively underexplored in prior literature. Trajectory data are collected from approximately 28 000 vehicles, resulting in 50 million movement traces, which are subsequently aggregated by scene to generate trajectory-modality images. The dataset contains 2011 image pairs, each covering urban areas within a total extent of 100 km^2, and annotated into two semantic classes: road and nonroad. All images share a uniform spatial resolution of 1024 × 1024 pixels. The data are divided into 70% for training, 10% for validation, and 20% for testing.

2) RS-RGB-T Dataset [54]: This dataset comprises urban road scenes captured by UAV-based RGB and thermal sensors, totaling 4024 paired optical–thermal images. Of these, 2514 are acquired during daytime and 1510 at night, with each image at a resolution of 640 × 512 pixels and annotated at the pixel level. The original annotations span nine semantic classes; for this study, only the road class is retained, with all other categories merged into the background. Dataset partitioning follows the source protocol, using a 7:1:2 split for training, validation, and testing, while maintaining the original day–night distribution in each subset.

3) Porto Dataset [55]: This dataset consists of paired optical and raw GPS trajectory bimodal data, with the low-quality sparse trajectory modality being relatively underexplored in existing multimodal road extraction studies. Trajectory data are sourced from the publicly available Porto Taxi Trajectory Dataset, covering 1.7 million raw taxi trip traces, with no smoothing, interpolation, denoising, or aggregation preprocessing applied to fully preserve the inherent sparsity and positional noise of raw GPS data, which are subsequently generated into trajectory-modality images by corresponding scene. The dataset contains 245 image pairs, covering urban areas within a total extent of 62.5 km^2, and annotated into two semantic classes: road and nonroad. All images share a uniform size of 1024 × 1024 pixels, where the optical images have a spatial resolution of 0.5 m. The dataset is divided into 70% for training, 10% for validation, and 20% for testing. It is specifically constructed to validate the robustness and generalization performance of the proposed model against low-quality and sparse GPS inputs.

B. Implementation Details

All experiments are conducted on a workstation equipped with an Intel Xeon Gold 6133 CPU (2.50 GHz), 24 GB RAM, and an NVIDIA GeForce RTX 3090 GPU, running Ubuntu 18.04. The proposed model is implemented in PyTorch 2.0.0. Network training is performed using the AdamW optimizer, which effectively mitigates overfitting and stabilizes gradient updates. Following the hyperparameter configuration of the original DINOv3 framework and further refinement via grid search, the model is trained for 50 epochs with an initial learning rate of 0.0001 and a weight decay of 0.01. Due to GPU memory constraints, the batch size is set to 1. To ensure reproducibility and statistical robustness, all results are reported as the average over three independent runs.

C. Evaluation Metrics

To comprehensively assess the performance of our model in multimodal road extraction, we adopt four standard semantic segmentation metrics: overall accuracy (OA), mean intersection over union (mIoU), mean $F1$ score (mF1), and the Kappa

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coefficient (κ). The OA measures the proportion of correctly classified pixels as follows:

$$\mathrm{OA} = \frac{\sum_{i=1}^{C} n_{ii}}{\sum_{i=1}^{C}\sum_{j=1}^{C} n_{ij}} \tag{37}$$

where n_{ij} denotes the number of pixels with ground truth i and predicted class j, and C is the number of classes. The mIoU evaluates the average overlap between prediction and ground truth across all classes

$$\mathrm{mIoU} = \frac{1}{C}\sum_{k=1}^{C} \frac{TP_k}{TP_k + FP_k + FN_k + \epsilon} \tag{38}$$

where TP_k, FP_k, and FN_k denote the true-positive, false-positive, and false-negative pixels for class k, respectively, and ϵ is a small constant for numerical stability. The mF1 provides a harmonic mean of precision and recall for each class

$$\mathrm{mF1} = \frac{1}{C}\sum_{i=1}^{C} \frac{2TP_i}{2TP_i + FP_i + FN_i}. \tag{39}$$

The Kappa coefficient (κ) quantifies the agreement between prediction and ground truth, corrected for chance

$$\kappa = \frac{\mathrm{OA} - \mathrm{Pe}}{1 - \mathrm{Pe}} \tag{40}$$

where $\mathrm{Pe} = \sum_{i=1}^{C}\left(\sum_{j=1}^{C} n_{ij}/N \cdot \sum_{j=1}^{C} n_{ji}/N\right)$ and N is the total number of pixels. Furthermore, we adopt APLS, a widely acknowledged topological evaluation metric in remote sensing road extraction. By conducting evaluations on the more challenging Porto dataset, we quantitatively assess the topological integrity and practical navigability of the extracted road networks. Specifically, APLS assesses the path connectivity and distance consistency between the predicted and ground-truth road networks, formulated as follows:

$$\mathrm{APLS} = \frac{1}{N}\sum_{n=1}^{N}\left[1 - \frac{\left|L_{\mathrm{pred}}(s_n, e_n) - L_{\mathrm{gt}}(s_n, e_n)\right|}{\max\left(L_{\mathrm{pred}}(s_n, e_n), L_{\mathrm{gt}}(s_n, e_n)\right)}\right] \tag{41}$$

where (s_n, e_n) denotes the shortest path length from the starting point s_n to the ending point e_n. These metrics comprehensively evaluate the pixel-level classification accuracy of the model while verifying its effectiveness in preserving road structure and maintaining topological consistency in complex multimodal remote sensing scenarios.

D. Comparison With State-of-the-Art Methods

To validate the effectiveness of the proposed DINO-MTP model, we conduct a comprehensive comparison against 16 state-of-the-art semantic segmentation approaches, including UNet [39], Road-SAM [43], RoadCT [42], DF-BSFNet [44], DECCFNet [10], CMIPNet [45], LCIRE-Net [46], MFNet [47], FlexiSAM [48], FTransUNet [49], MultiSenseSeg [50], SACFNet [51], DEFNet [26], RingMo-SAM [52], LMGANet [56], and CFIFNet [57], evaluated on three benchmark datasets. Detailed comparative results and ablation analyses are presented in Section IV-D1–IV-D3.

1) Results on BJRoad Dataset: To objectively evaluate the performance of DINO-MTP, we benchmark it against 16 state-of-the-art segmentation methods on the BJRoad dataset. As summarized in Table I, DINO-MTP achieves an mF1 of 78.62% and an mIoU of 71.67%, outperforming the strongest prior multimodal baseline, LMGANet, by 1.06% and 1.92%, respectively. In terms of OA and agreement, DINO-MTP achieves the best OA of 87.06%, outperforming LMGANet by 0.70%. For Kappa, DINO-MTP obtains 81.62%, which is higher than CFIFNet by 0.80%. These results reflect improved pixel-level classification.

Qualitative results in Fig. 4 further corroborate the superiority of DINO-MTP across diverse and challenging urban scenarios. The proposed model produces complete and continuous road masks with sharp boundaries and robust global connectivity. In dense urban blocks, DINO-MTP accurately separates adjacent roads and preserves intricate intersection structures. For narrow, curved, and secondary roads, DINO-MTP demonstrates heightened sensitivity, effectively capturing slender and tortuous segments that are frequently missed or fragmented by competing methods. Moreover, in visually ambiguous or weak-texture regions, as well as areas with sparse trajectory data or occlusions, DINO-MTP exhibits strong resilience, minimizing omissions and false connections. By contrast, existing multimodal baselines (e.g., RingMo-SAM, LMGANet, and CFIFNet) generally perform well on prominent roadways but tend to suffer from structural discontinuities, local breaks, and large omission regions, especially under occlusions, shadows, or limited trajectory coverage. They are also more susceptible to boundary blurring and false positives in cluttered backgrounds. The marked improvements of DINO-MTP are mainly attributable to the multimodal feature fusion module, which enhances semantic alignment and noise robustness while preserving road network topology. By generating stable, well-aligned multimodal representations, this module lays a reliable foundation for spatial-prior modeling and structural reconstruction, bridging the domain gap between optical imagery and trajectory data. Overall, both quantitative and qualitative results on BJRoad demonstrate that DINO-MTP not only excels in fine-grained road structure extraction but also exhibits strong generalization and applicability for multimodal road extraction in complex urban environments.

2) Results on RS-RGB-T Dataset: The RS-RGB-T dataset with RGB and thermal infrared modalities enables comprehensive evaluation of model generalization and robustness in complex urban scenes, especially under low illumination, occlusion, or clutter. As shown in Table II, DINO-MTP achieves state-of-the-art performance across all evaluation metrics, including OA, mF1, mIoU, and Kappa, substantially outperforming both unimodal and multimodal baselines. Specifically, DINO-MTP attains an mF1 of 94.62% and an mIoU of 89.65%, exceeding LCIRE-Net and LMGANet by 0.33% and 1.38%, respectively. For OA and agreement, DINO-MTP reaches 97.96% in OA and 96.57% in Kappa, with improvements of 0.14% and 0.11% over DEFNet and CFIFNet, respectively. Qualitative results in Fig. 5 further demonstrate that DINO-MTP consistently preserves road

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TABLE I

PERFORMANCE COMPARISON OF DINO-MTP AND 16 STATE-OF-THE-ART SEGMENTATION METHODS ON THE BJROAD DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Type	Methods	OA	mF1	mIoU	Kappa
UniModal	UNet [41]	71.38	64.18	59.42	70.05
	RoadCT [44]	79.68	70.42	60.91	75.36
	Road-SAM [45]	82.55	70.33	62.80	74.50
Multimodal	DF-BSFNet [46]	81.94	71.86	61.64	81.57
	DECCFNet [47]	80.71	71.46	61.83	77.35
	CMIPNet [48]	80.47	70.35	61.04	79.77
	LCIRE-Net [49]	78.94	69.44	62.57	76.09
	MFNet [50]	80.15	72.98	65.19	77.14
	FlexiSAM [51]	83.12	74.65	63.37	79.80
	FTransUNet [52]	83.87	76.54	65.16	79.80
	MultiSenseSeg [53]	83.35	77.07	66.17	77.59
	SACFNet [54]	84.36	74.11	63.44	79.10
	DEFNet [55]	83.41	75.50	64.18	79.74
	RingMo-SAM [56]	86.13	75.32	65.07	80.37
	LMGANet [60]	86.36	77.56	69.75	81.85
	CFIFNet [61]	86.05	76.96	68.74	80.82
	DINO-MTP(Ours)	**87.06**	**78.62**	**71.67**	**81.62**

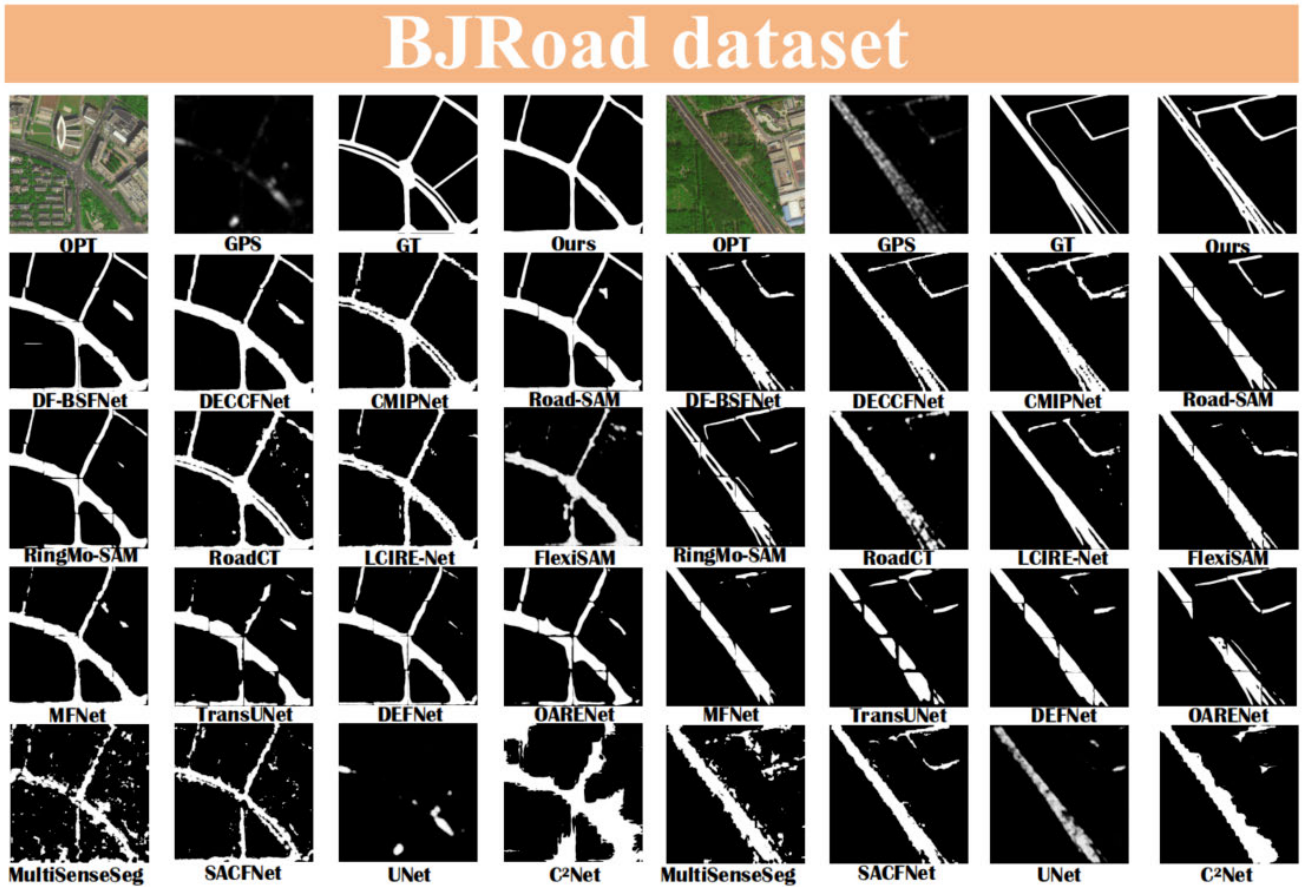


Fig. 4. Qualitative comparison of road extraction by various methods on the BJRoad dataset in complex urban scenes.

connectivity, accurately recovers weak-texture and occluded segments, and maintains high geometric fidelity across diverse urban scenarios. Predicted road masks are smoother, more complete, and less prone to fragmentation. In contrast, existing methods, such as LMGANet, DEFNet, and CFIFNet, tend to miss narrow roads or generate fragmented predictions, particularly in areas affected by occlusion, thermal variation, or weak optical signals. In dense urban scenes, models such as MultiSenseSeg, SACFNet, and RingMo-SAM are susceptible to misclassifying nonroad regions, such as parking lots or building shadows, resulting in irregular artifacts and increased class confusion. The superior performance of DINO-MTP can

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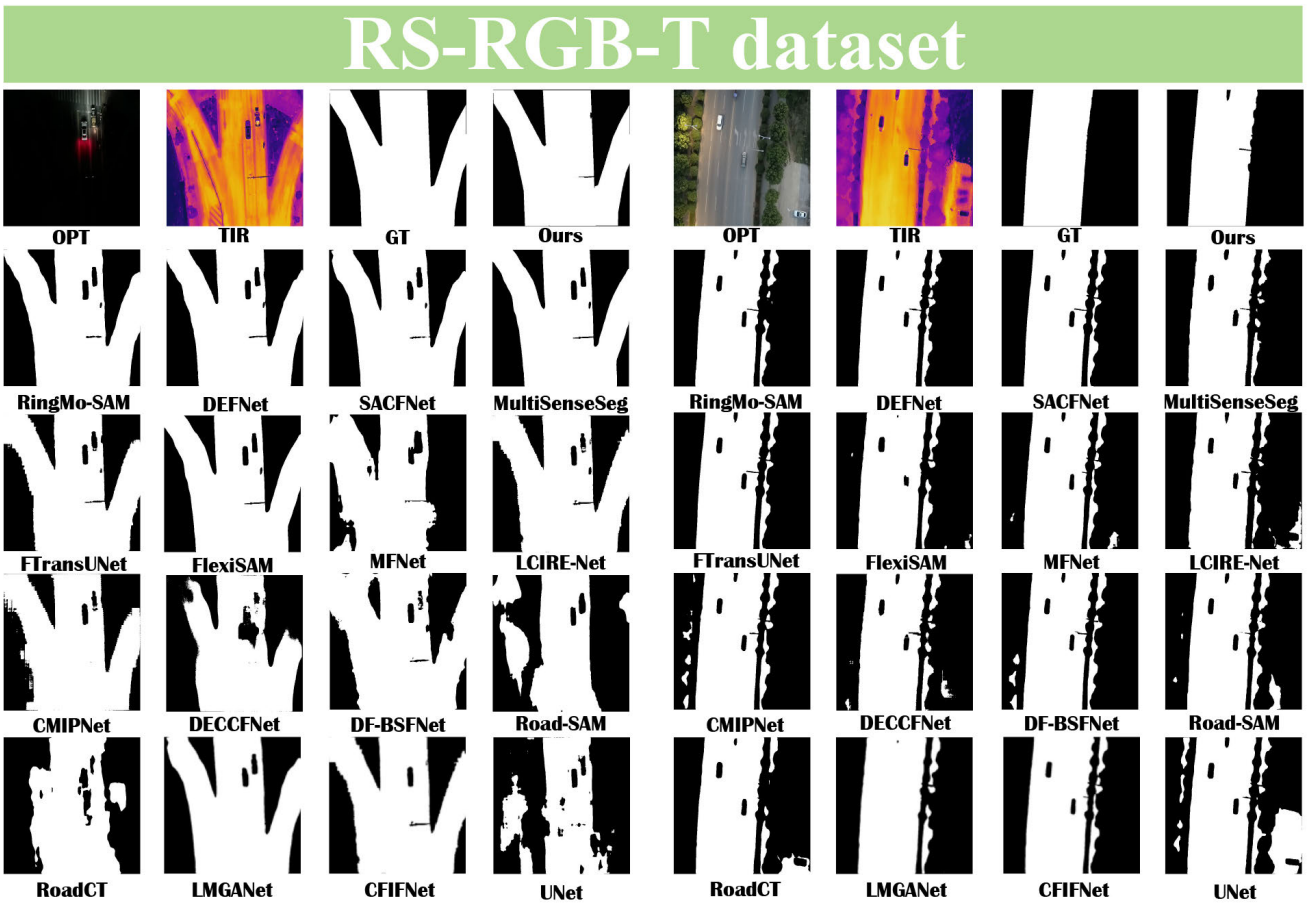


Fig. 5. Qualitative comparison of road extraction by various methods on the RS-RGB-T dataset in complex urban scenes.

TABLE II

PERFORMANCE COMPARISON OF DINO-MTP AND 16 STATE-OF-THE-ART SEGMENTATION METHODS ON THE RS-RGB-T DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Type	Methods	OA	mF1	mIoU	Kappa
UniModal	UNet [41]	82.41	76.20	73.18	79.60
	RoadCT [44]	85.47	80.23	75.12	83.45
	Road-SAM [45]	89.89	82.43	80.06	88.92
Multimodal	DF-BSFNet [46]	96.93	90.48	86.19	96.79
	DECCFNet [47]	96.54	89.88	85.74	90.84
	CMIPNet [48]	93.57	90.77	85.42	93.21
	LCIRE-Net [49]	96.20	94.29	88.22	94.88
	MFNet [50]	94.50	89.93	83.19	90.91
	FlexiSAM [51]	94.59	90.22	87.48	91.28
	FTransUNet [52]	96.12	93.47	86.91	94.83
	MultiSenseSeg [53]	95.37	91.41	87.51	94.18
	SACFNet [54]	95.72	92.84	86.61	93.29
	DEFNet [55]	97.82	92.47	87.23	94.16
	RingMo-SAM [56]	94.19	93.08	86.58	93.50
	LMGANet [60]	96.95	93.34	88.27	95.64
	CFIFNet [61]	97.41	93.95	87.64	96.46
	DINO-MTP(Ours)	**97.96**	**94.62**	**89.65**	**96.57**

be attributed to the introduction of SPMCA, which integrates local–global spatial priors with multiscale statistical enhancement.

This module strengthens the representation of linear and ribbon-like road structures while effectively suppressing high-frequency noise and cross-modal inconsistencies. Consequently, the fused multimodal features remain robust and sensitive to fine-scale structures, supporting accurate reconstruction of narrow roads, curved segments, and complex intersections. From these results, DINO-MTP demonstrates

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TABLE III
PERFORMANCE COMPARISON OF DINO-MTP AND 16 STATE-OF-THE-ART SEGMENTATION METHODS ON THE PORTO DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Type	Methods	OA	mF1	mIoU	Kappa	APLS (↑)
UniModal	UNet [41]	63.15	67.14	60.35	59.21	52.36
	RoadCT [44]	64.35	69.47	65.13	61.33	55.19
	Road-SAM [45]	65.78	70.95	66.68	63.19	57.28
Multimodal	DF-BSFNet [46]	68.98	72.38	68.35	67.47	60.67
	DECCFNet [47]	69.46	73.96	69.16	67.92	59.34
	CMIPNet [48]	71.65	76.24	70.84	70.29	62.85
	LCIRE-Net [49]	72.39	75.12	72.95	70.91	63.74
	MFNet [50]	73.28	76.35	73.16	71.54	64.59
	FlexiSAM [51]	73.63	78.12	76.10	71.82	65.71
	FTransUNet [52]	75.36	80.63	74.19	73.91	68.24
	MultiSenseSeg [53]	76.92	83.62	75.64	74.76	71.64
	SACFNet [54]	80.09	81.97	76.95	77.96	70.53
	DEFNet [55]	79.56	86.19	77.15	77.48	72.10
	RingMo-SAM [56]	80.62	85.61	78.06	78.18	73.52
	LMGANet [60]	80.94	86.37	78.34	78.27	74.11
	CFIFNet [61]	79.41	86.98	77.18	77.45	73.38
	DINO-MTP(Ours)	**81.32**	**88.45**	**78.89**	**79.24**	**75.08**

strong generalization and reliability on the RS-RGB-T dataset, establishing its effectiveness for high-precision multimodal road extraction in urban environments.

3) Results on Porto Dataset: On the Porto dataset for complex urban road networks, as shown in Table III, DINO-MTP achieves comprehensive superiority across all evaluation metrics. Specifically, in terms of pixel-level classification accuracy, our method obtains an OA of 81.32%, an mF1 of 88.45%, and an mIoU of 78.89%. Compared to the best-performing suboptimal multimodal baseline models, CFIFNet and LMGANet, DINO-MTP yields improvements of 1.47% and 0.55% in mF1 and mIoU, respectively, while reaching the highest Kappa coefficient of 79.24%. This fully demonstrates that through the effective synergy of SPMCA and FAPMM, our method explicitly enhances the sensitivity to elongated tubular structures during both the encoding and decoding stages, and exhibits exceptional robustness in distinguishing roads from complex backgrounds and overcoming class imbalance. Besides conventional pixel-level metrics, DINO-MTP achieves a significant improvement in APLS to 75.08, demonstrating that our method considerably outperforms other comparative approaches in maintaining the global structural integrity of road networks. This significant graph-level quantitative advantage is further corroborated by the qualitative visualization results in Fig. 6. As illustrated in Fig. 6, traditional unimodal methods, such as UNet and RoadCT, rely heavily on single optical features, making them highly susceptible to severe road disconnections and large-scale omission errors in areas with shadow occlusions or weak textures. Existing multimodal methods, such as DF-BSFNet and CMIPNet, although incorporating trajectory priors, still suffer from obvious fragmentation phenomena at fine branches and complex intersections. Even CFIFNet and LMGANet, which possess higher quantitative metrics, remain inadequate in maintaining the connectivity of local minor road segments, generating partial artifacts and disconnections. In contrast, by effectively aligning the fine textures of optical images with the global skeleton priors of GPS trajectories via the MDFFM module, DINO-MTP significantly filters out the artifacts caused by sparse GPS data. The extracted road network not only features extremely clear boundaries but also demonstrates exceptionally high topological fidelity visually. Based on the aforementioned results, DINO-MTP demonstrates outstanding topology-preserving capability and robustness on the Porto dataset. It establishes its effectiveness for high-connectivity multimodal road extraction in complex urban environments, providing a crucial solution to overcome the bottlenecks of road network disconnections and fragmentation that frequently occur in traditional road extraction tasks.

E. Ablation Studies

To rigorously evaluate the contribution of each core component in DINO-MTP to multimodal road extraction and spatial topology reconstruction, we conduct extensive ablation experiments on both the BJRoad and RS-RGB-T datasets. Using the baseline framework as a reference, we sequentially introduce the following key modules: MDFFM, SPMCA, FAPMM, and DPCF-ECA, while keeping the training settings, data splits, and hyperparameters fixed.

1) Effectiveness of MDFFM: The effectiveness of the MDFFM is evidenced by consistent improvements in both quantitative metrics and qualitative performance. As shown in Table IV, incorporating MDFFM into the baseline yields substantial gains: on BJRoad, mF1 increases from 74.23% to 76.56% and mIoU from 67.33% to 69.02%; on RS-RGB-T, mF1 rises from 90.33% to 92.32% and mIoU from 86.28% to 87.85%. Similar trends are observed when MDFFM is integrated into different module combinations, further validating its generality and effectiveness. Feature visualizations in Fig. 7 show that MDFFM significantly suppresses irrelevant activations in challenging regions, such as shadows and pseudotrajectory areas, resulting in more focused and discriminative feature responses. As illustrated in Fig. 8, the addition of MDFFM leads to more complete and accurate

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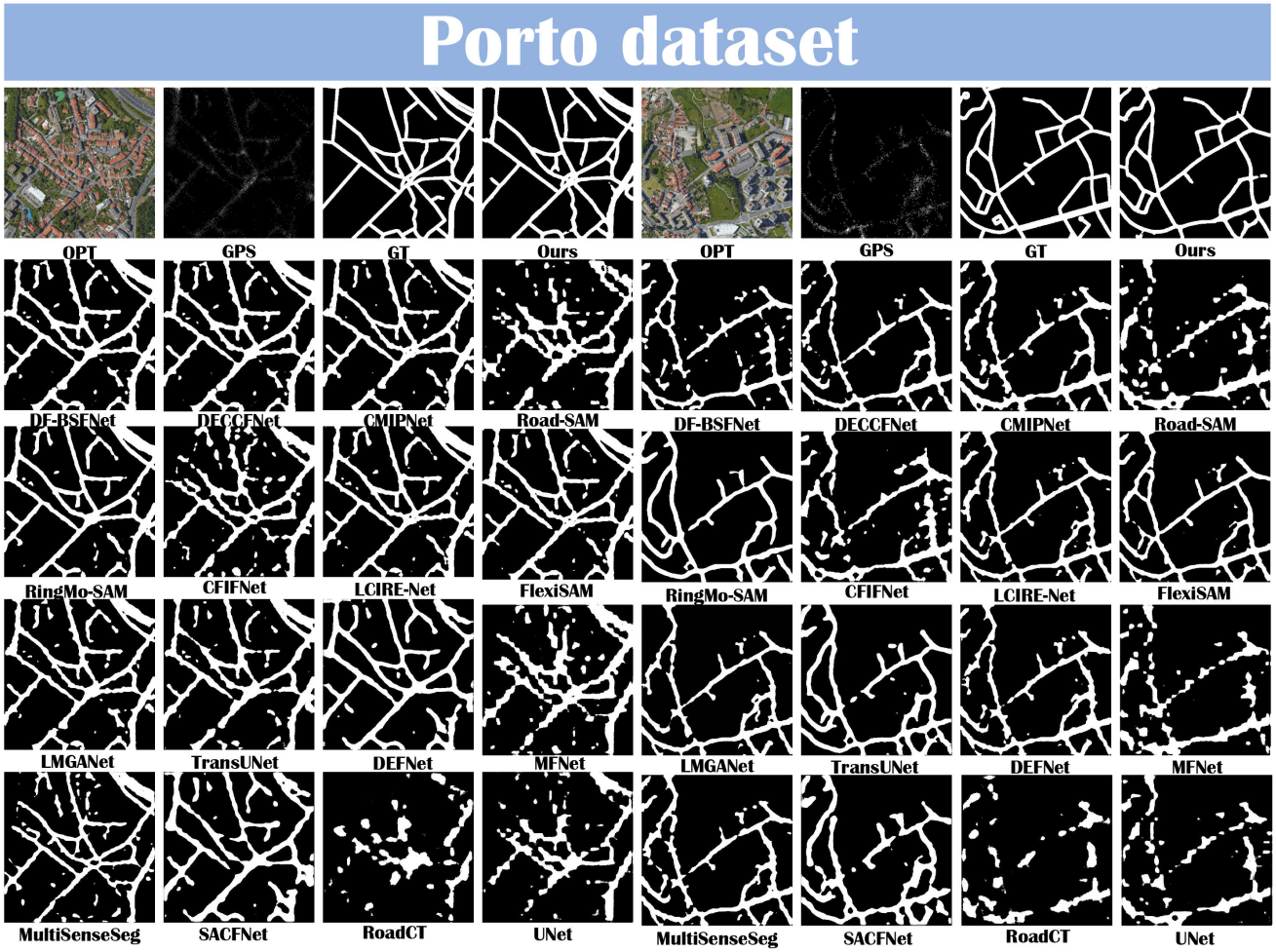


Fig. 6. Qualitative comparison of road extraction by various methods on the Porto dataset in complex urban scenes.

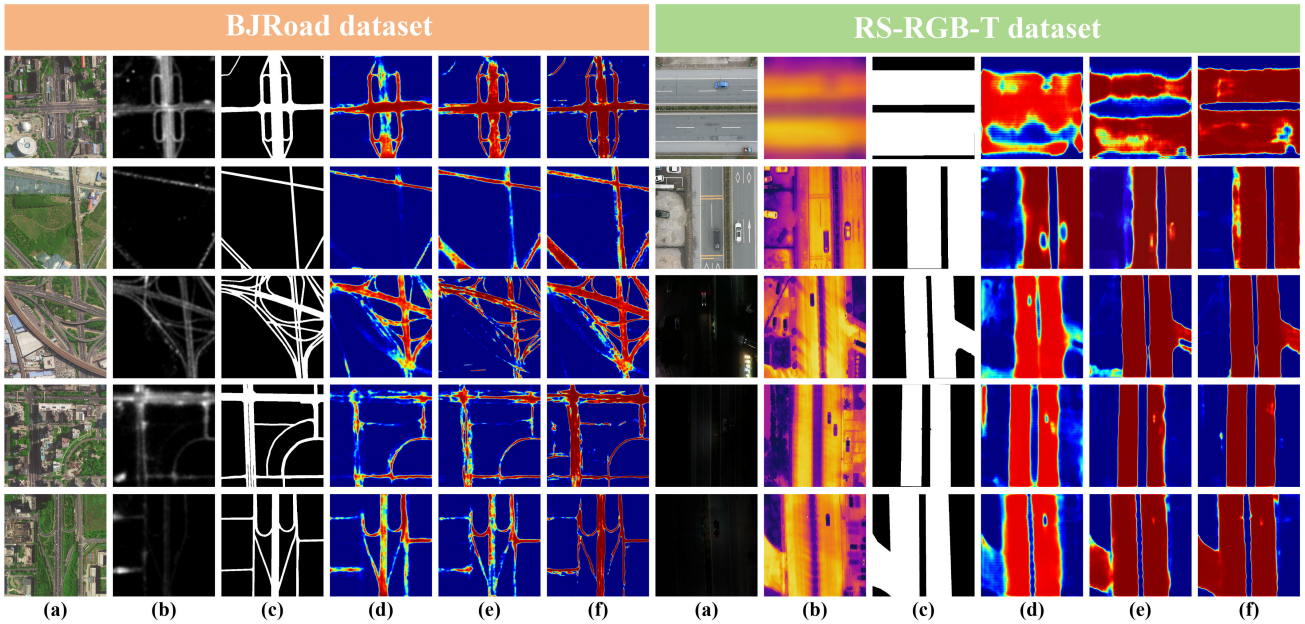


Fig. 7. Feature visualizations of different module combinations on BJRoad and RS-RGB-T datasets. (a) OPT. (b) GPS/TIR. (c) GT. (d) +MDFFM. (e) +MDFFM+SPMCA. (f) +MDFFM+SPMCA+FAPMM/DPCF-ECA.

road extraction, with improved connectivity and fewer false positives. This is further supported by the progressive visual ablation in Fig. 9. Specifically, MDFFM [see Fig. 9(e)] significantly suppresses background noise and false positives in the baseline [see Fig. 9(d)], especially in complex scenes of the first and 4th rows. Furthermore, t-SNE distributions

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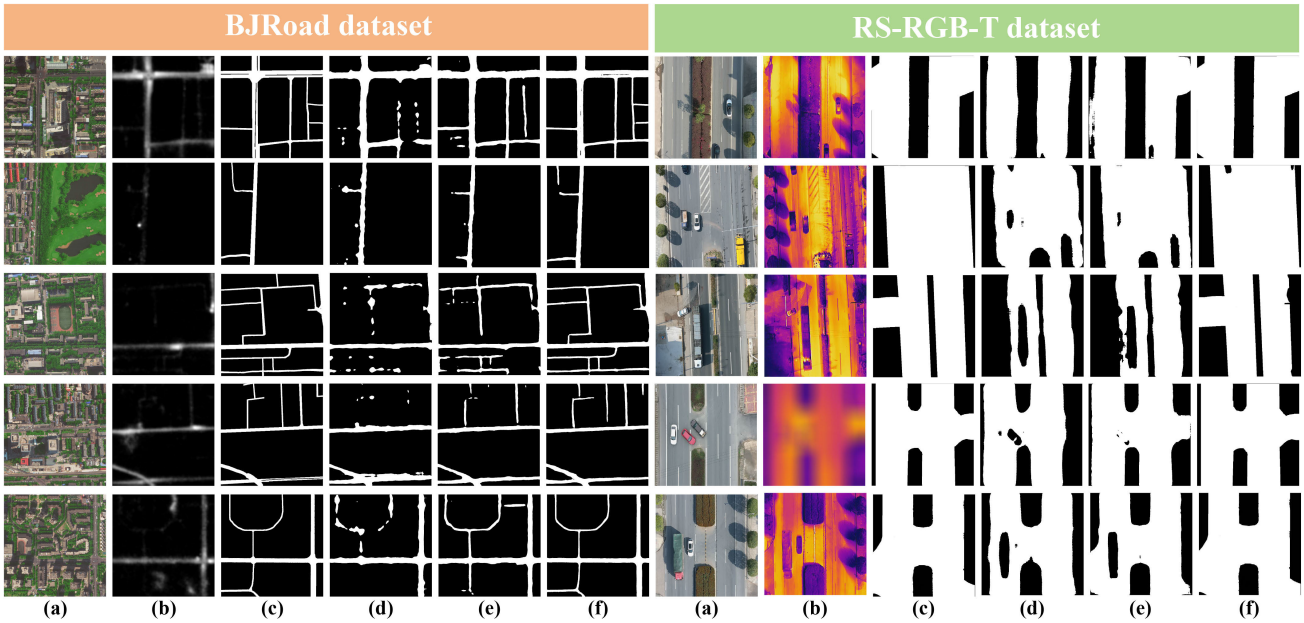


Fig. 8. Qualitative comparison of road extraction results for various ablation settings on BJRoad and RS-RGB-T datasets. (a) OPT. (b) GPS/TIR. (c) GT. (d) +MDFFM. (e) +MDFFM+SPMCA. (f) +MDFFM+SPMCA+FAPMM/DPCF-ECA.

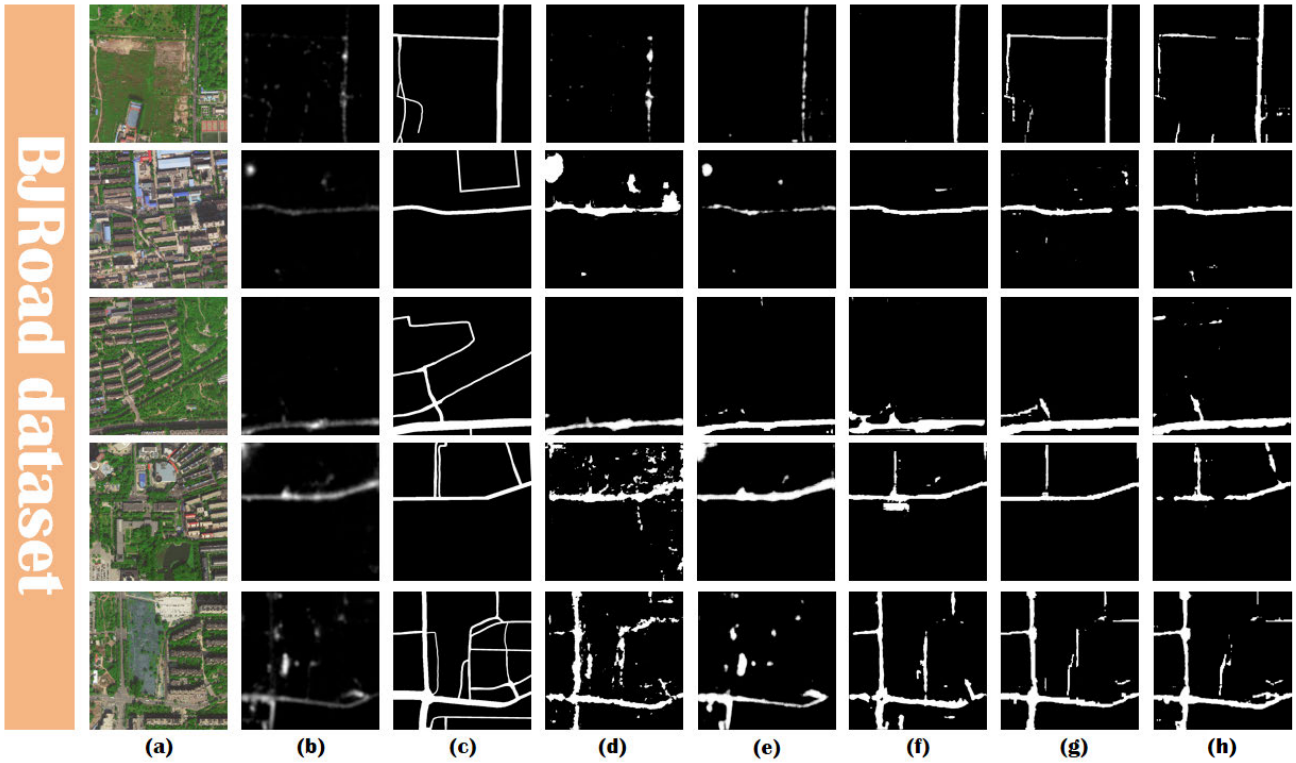


Fig. 9. Qualitative comparison of road extraction results for various ablation settings on BJRoad dataset. (a) OPT. (b) GPS. (c) GT. (d) Baseline. (e) +MDFFM. (f) +SPMCA. (g) +FAPMM. (h) +DPCF-ECA.

in Fig. 10 reveal that MDFFM enhances the separation between road and background clusters, indicating improved feature discriminability and noise suppression. Collectively, these results demonstrate that MDFFM effectively addresses modality misalignment and background interference, thereby facilitating robust multimodal fusion and enhancing salient road feature extraction.

2) Effectiveness of the SPMCA: Building upon the fused representations, SPMCA further improves the continuity and integrity of thin and complex road structures. As shown in Table IV, incorporating SPMCA brings consistent and notable quantitative gains across multiple metrics and datasets. On BJRoad, the addition of SPMCA increases the mIoU from 69.02% to 70.18%, and the mF1 from 76.56% to 77.89%.

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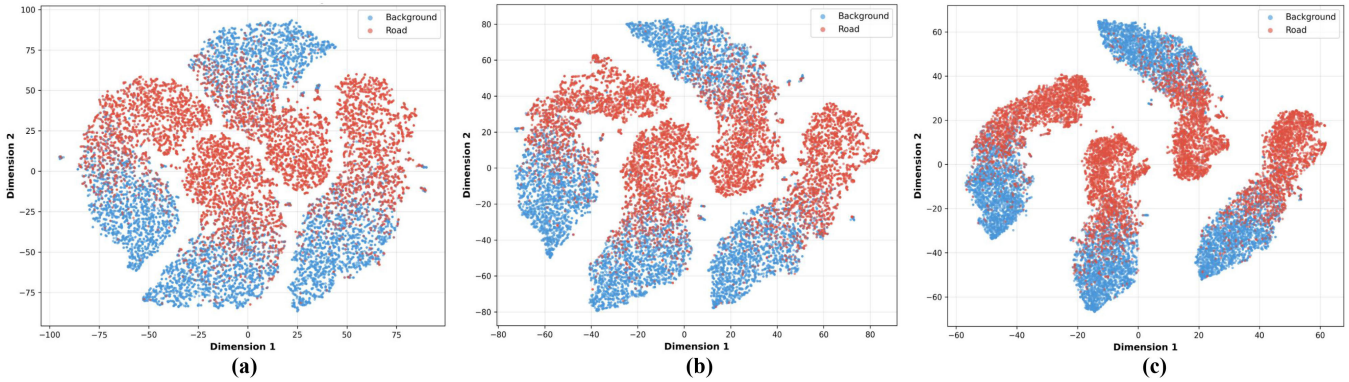


Fig. 10. t-SNE visualization of feature distributions. (a) +MDFFM. (b) +MDFFM+SPMCA. (c) +MDFFM+SPMCA+FAPMM/DPCF-ECA.

TABLE IV

ABLATION STUDY OF THE PROPOSED MODULES ON THE BJROAD AND RS-RGB-T DATASETS. EACH MODULE IS INCREMENTALLY ADDED TO THE BASELINE TO ASSESS ITS INDIVIDUAL CONTRIBUTION

Models	MDFFM	SPMCA	FAPMM DPCF-ECA	BJRoad			RS-RGB-T		
				OA	mF1	mIoU	OA	mF1	mIoU
Baseline	×	×	×	83.69	74.23	67.33	93.96	90.33	86.28
DINO-MTP	✓	×	×	85.69	76.56	69.02	95.45	92.32	87.85
	×	✓	×	86.13	77.89	70.18	96.53	93.55	88.03
	×	×	✓	86.52	78.15	70.63	97.11	93.96	88.75
	✓	✓	×	84.78	75.19	68.23	95.58	92.45	86.69
	✓	×	✓	85.19	76.56	69.89	96.05	93.32	87.85
	×	✓	✓	85.89	78.21	70.36	96.88	94.02	89.13
	✓	✓	✓	87.06	78.62	71.67	97.96	94.62	89.65

Similar improvements are observed on RS-RGB-T, where mIoU rises from 87.85% to 88.03%, and mF1 from 92.32% to 93.55%. Notably, SPMCA also brings further gains when integrated into the full model, with the best configuration achieving 71.67% mIoU and 78.62% mF1 on BJRoad, and 89.65% mIoU on RS-RGB-T. In addition to quantitative results, qualitative visualizations in Fig. 7 show that SPMCA produces stronger, more continuous activations along narrow roads and more complete intersection extraction. Fig. 8 demonstrates that SPMCA yields finer and more coherent road extraction results, reducing fragmentation. As explicitly demonstrated in the progressive ablation results illustrated in Fig. 9(f), benefiting from the embedded spatial priors, SPMCA effectively reconnects the broken road segments generated in the previous stage, as typified in the fourth and fifth rows, substantially mitigating the fragmentation issue and enhancing topological continuity. The t-SNE feature distributions in Fig. 10 further highlight improved class separability, with sharper boundaries between road and background clusters. These results verify that SPMCA explicitly encodes spatial priors, thereby significantly improving the extraction of continuous and fine-grained road structures.

3) Effectiveness of the FAPMM and DPCF-ECA: To fully leverage cross-scale geometric cues, we jointly ablate FAPMM (encoder) and DPCF-ECA (decoder), as these modules are complementary: FAPMM enhances fine-grained, high-resolution representations, while DPCF-ECA adaptively fuses multiscale details with global topology. Grouping them highlights their combined effect on geometric completeness and detail preservation at different network stages. As shown in Table IV, the integration of FAPMM and DPCF-ECA leads to the highest scores across all metrics and datasets. For instance, on BJRoad, the full model achieves an mF1 of 78.62% and mIoU of 71.67%, representing improvements of +4.39% and +4.34% over the baseline, respectively. On RS-RGB-T, the mIoU reaches 89.65% and mF1 94.62%, with corresponding OA increases as well. These gains are not only consistent across datasets but also pronounced in challenging scenarios involving occluded or narrow road segments. Qualitative results further support these findings. The feature visualizations in Fig. 7 demonstrate that the joint modules yield stronger responses along slender and ambiguous roads, with more complete restoration of intersections and long-range structures. As shown in Fig. 8, road extraction results exhibit fewer breaks and missing segments, and improved topological integrity. The progressive comparisons in Fig. 9(g) and (h) visually verify the joint effectiveness. By sequentially applying FAPMM and DPCF-ECA, road boundaries are refined and broken topologies are restored, yielding a continuous, topologically complete road network highly consistent with the ground truth. Meanwhile, the t-SNE plots in Fig. 10 indicate that the extracted features are more compact within classes and better separated between road and background.

F. Parameter Sensitivity Analysis

To address the core challenges in high-resolution remote sensing image road extraction, including class imbalance, road

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TABLE V
PARAMETER SENSITIVITY ANALYSIS OF THE COMPOSITE LOSS FUNCTION ON THE BJROAD DATASET

dataset	λ_1	λ_2	λ_3	OA	mF1	mIoU	Kappa
BJRoad	0.4	0.3	0.3	85.97	79.87	71.51	80.39
	0.3	0.6	0.1	86.54	78.21	72.62	80.94
	0.3	0.1	0.6	87.10	77.37	70.28	81.73
	0.8	0.1	0.1	85.36	78.51	70.69	81.49
	0.6	**0.3**	**0.1**	**87.06**	**78.62**	**71.67**	**81.62**

TABLE VI
COMPUTATIONAL EFFICIENCY COMPARISON OF DIFFERENT METHODS

Modality	Method	FLOPs	Params	Mem.	Speed
UniModal	UNet	120.8	62.5	2280	82
	OARENet	111.5	62.1	2390	78
	C2Net	94.5	68.2	2511	69
	RoadCT	98.5	64.9	2455	72
	Road-SAM	93.2	70.4	2588	62
Multimodal	DF-BSFNet	85.2	82.1	2954	52
	DECCFNet	90.1	78.3	2701	55
	CMIPNet	46.6	130.1	4108	28
	LCIRE-Net	86.9	64.5	2512	62
	MFNet	85.1	66.1	2669	56
	FlexiSAM	50.8	124.2	4169	30
	FTransUNet	48.5	102.8	3571	33
	MultiSenseSeg	53.0	94.8	3169	32
	SACFNet	83.4	69.3	2418	49
	DEFNet	72.4	80.6	2890	48
	RingMo-SAM	58.3	108.4	3423	35
	DINO-MTP	71.7	93.4	4375	51

topological discontinuity, and inaccurate boundary localization, this article designs a composite loss function expressed as follows:

$$L_{\text{total}} = \lambda_1 L_{\text{WCE}} + \lambda_2 L_{\text{GDL}} + \lambda_3 L_{\text{BAL}}. \tag{42}$$

To determine the optimal weight combination, a parameter sensitivity analysis is conducted on the BJRoad dataset, using four key metrics (mIoU, OA, Kappa, and mF1) to quantify extraction performance (see Table V). When $\lambda_1 = 0.3$, $\lambda_2 = 0.1$, and $\lambda_3 = 0.6$, the model achieves the highest OA and Kappa, but the core mIoU drops to 70.28%, as excessive focus on boundary optimization sacrifices the integrity of road network topology. When $\lambda_1 = 0.3$, $\lambda_2 = 0.6$, and $\lambda_3 = 0.1$, the mIoU reaches the maximum of 72.62%, but the mF1 decreases to 78.21% due to the neglect of boundary refinement. After a comprehensive comparison, the optimal weight configuration is determined as $\lambda_1 = 0.6$, $\lambda_2 = 0.3$, and $\lambda_3 = 0.1$. Under this configuration, the model achieves a balance between pixel-level classification accuracy, road network connectivity, and boundary precision, fully meeting the requirements of high-precision road extraction from high-resolution remote sensing images.

G. Computational Complexity Analysis

To comprehensively assess computational efficiency, we compare DINO-MTP with state-of-the-art unimodal and multimodal methods in terms of FLOPs, parameter count, memory usage, and inference speed (FPS), as shown in Table VI. All methods are evaluated under identical hardware and input settings for fairness. Unimodal baselines, like UNet and OARENet, have high computational costs, with FLOPs exceeding 110 G and parameter counts over 60 M, though their inference speeds remain relatively high (62–82 FPS). Multimodal methods generally require more resources due to complex fusion strategies. For example, CMIPNet and FlexiSAM, despite moderate FLOPs, have high memory and parameter demands, resulting in lower speeds. DINO-MTP achieves a balanced tradeoff, with 71.7 G FLOPs and 93.4 M parameters, achieving a practical tradeoff between accuracy and computational efficiency among multimodal methods. DINO-MTP requires the highest memory consumption among the compared methods due to the dual DINOv3 backbone. Nevertheless, it maintains a competitive inference speed of 51 FPS and moderate computational complexity in terms of FLOPs compared with several recent multimodal baselines. During training, although the dual backbone requires a peak memory of approximately 16.2 GB and 18 min per epoch, it converges significantly faster due to robust pretrained weights. Furthermore, DINO-MTP demonstrates strong scalability for high-resolution imagery, as the patch-based mechanism of DINOv3 and groupwise computation in DPCF-ECA effectively manage memory constraints during massive tile processing. These results underscore the efficient design of DINO-MTP, which delivers strong segmentation accuracy without sacrificing practical deployability.

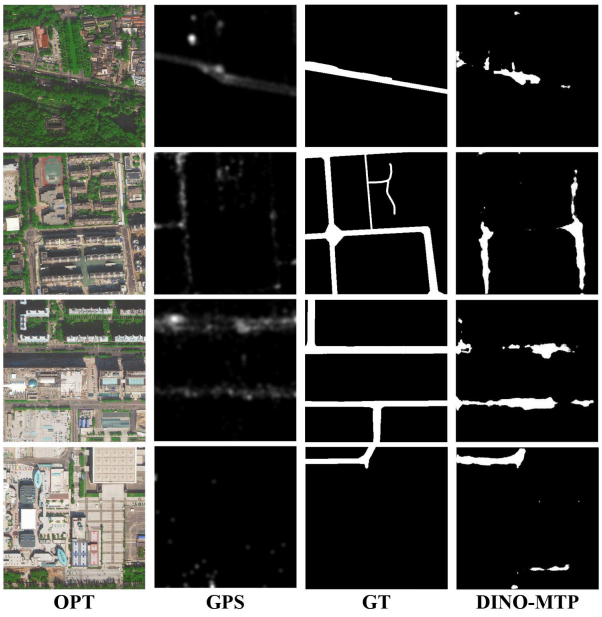


Fig. 11. Representative failure cases of DINO-MTP in complex urban scenes.

H. Failure Case Analysis

Although DINO-MTP achieves strong overall results in multimodal road extraction, several challenging cases remain, particularly in visually complex urban environments. As illustrated in Fig. 11, the model may misclassify large paved areas or open spaces in residential neighborhoods as roads, leading

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to boundary overexpansion or spurious connections between nonroad regions. In riverfront and expressway scenes, riverside walkways or parking lots are sometimes confused with main roadways, while true lanes obscured by trees tend to appear fragmented or discontinuous. In large urban complexes with plazas and open pavement, high-reflectance edges or interior regions are often mistakenly labeled as roads, producing block-shaped false positives, whereas narrow access roads may be only partially extracted. These cases suggest that DINO-MTP still faces difficulties in distinguishing visually similar but semantically different regions, capturing fine-grained contextual cues for accurate boundaries, and preserving network topology under occlusion or high scene heterogeneity. Future improvements could focus on enhancing contextual reasoning, leveraging stronger topological priors and uncertainty modeling to further improve robustness and generalization.

V. Conclusion

In this work, we proposed DINO-MTP, a novel topology-aware multimodal fusion framework for remote sensing road extraction that effectively integrates GPS trajectory data and optical imagery. By leveraging a self-supervised DINOv3 backbone and introducing modules, such as the MDFFM, SPMCA, FAPMM, and DPCF-ECA, our approach explicitly addresses cross-modal semantic alignment, structural discontinuities, and fine-grained boundary recovery. Extensive experiments on three challenging multimodal benchmarks, BJRoad, RS-RGB-T, and Porto, demonstrate that DINO-MTP consistently outperforms a wide range of state-of-the-art unimodal and multimodal baselines in both accuracy and topological integrity, especially under complex and heterogeneous urban scenarios. Ablation studies further validate the effectiveness of each proposed module in enhancing multimodal feature representation, structural detail preservation, and network robustness. The framework also achieves a favorable balance between computational efficiency and segmentation performance, supporting its practical deployment in large-scale urban mapping applications. To improve our method's robustness, generalization, and applicability, future work will focus on three directions: extending DINO-MTP to other linear-structure extraction tasks (e.g., rivers, railways, and power lines) with similar topological constraints; developing uncertainty-aware mechanisms for extreme conditions with noisy or missing GPS data; and integrating topology-aware fusion modules into remote sensing foundation models (e.g., SpectralGPT and RingMo) for zero-shot multimodal capabilities.

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